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TREND-BASED MOCK PAPER

GATE Physics

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Physics
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

$5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Section B: Physics

Q2. [1-NAT] [1 mark]

d/dx of e^x :

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q4. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q5. [1-NAT] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
 - (B) 0.092
 - (C) 0.762
 - (D) 0.195
-

Q7. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
 - (B) e^2
 - (C) 1
 - (D) infinity
-

Q8. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
 - (B) -2
 - (C) 1
 - (D) 2
-

Q9. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
 - (B) Differentiable only
 - (C) Continuous on $[a,b]$ and differentiable on (a,b)
 - (D) Both continuous and differentiable everywhere
-

Q10. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{\|u\| * \|v\|}$
-

Q11. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q13. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
 - (B) Cosine terms
 - (C) Both
 - (D) Neither
-

Q15. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q16. [2-MSQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q17. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1,0]^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-NAT] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
 - (B) 0.5
 - (C) 0.6
 - (D) 0.4
-

Q21. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q22. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q23. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) $e^{-3/s}$

Q24. [2-NAT] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MSQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q26. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q27. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q29. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q30. [1-MCQ] [1 mark]

Electric flux =

- (A) $E \cdot dA$
- (B) q/ϵ_0
- (C) Both
- (D) None

Q31. [1-MCQ] [1 mark]

Boundary condition: tangential E continuous across:

- (A) Dielectric boundary
- (B) Conductor surface
- (C) Both
- (D) Neither

Q32. [1-MCQ] [1 mark]

Maxwell added displacement current to:

- (A) Faraday
- (B) Ampere
- (C) Gauss
- (D) Ohm

Q33. [2-MCQ] [2 marks]

Plane wave in free space: E and H are:

- (A) Parallel
- (B) Perpendicular
- (C) Anti-parallel
- (D) Same direction

Q34. [2-NAT] [2 marks]

Poynting vector represents:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-NAT] [2 marks]

Maxwell added displacement current to:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MCQ] [2 marks]

Plane wave in free space: E and H are:

- (A) Parallel
- (B) Perpendicular
- (C) Anti-parallel
- (D) Same direction

Q37. [2-MCQ] [2 marks]

Maxwell added displacement current to:

- (A) Faraday
- (B) Ampere
- (C) Gauss
- (D) Ohm

Q38. [1-NAT] [1 mark]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

(Numerical Answer Type — Enter the numerical value)

Q39. [1-MCQ] [1 mark]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
 - (B) r and γ
 - (C) Temperature only
 - (D) Pressure only
-

Q40. [1-MCQ] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

- (A) 500J
 - (B) 333J
 - (C) 667J
 - (D) 1000J
-

Q41. [1-MCQ] [1 mark]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

- (A) 1729 J
 - (B) 0 J
 - (C) -1729 J
 - (D) -864 J
-

Q42. [1-MCQ] [1 mark]

Entropy change of universe for reversible process:

- (A) >0
 - (B) $=0$
 - (C) <0
 - (D) Depends
-

Q43. [2-MCQ] [2 marks]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
 - (B) 50%
 - (C) 60%
 - (D) 80%
-

Q44. [2-MCQ] [2 marks]

$C_p - C_v$ for ideal gas =

- (A) 0
 - (B) R
 - (C) $2R$
 - (D) $R/2$
-

Q45. [2-NAT] [2 marks]

Max efficiency of Carnot between 500K and 300K:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-NAT] [1 mark]

Laser principle:

(Numerical Answer Type — Enter the numerical value)

Q47. [1-MCQ] [1 mark]

Young's double slit: fringe width $\beta =$

- (A) $\lambda L/d$
 - (B) $\lambda d/L$
 - (C) $L/(\lambda d)$
 - (D) $d/(\lambda L)$
-

Q48. [1-NAT] [1 mark]

Laser principle is based on:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Numerical aperture of optical fiber:

- (A) $\sin(\theta_c)$
 - (B) $\sqrt{n_1^2 - n_2^2}$
 - (C) n_2/n_1
 - (D) n_1/n_2
-

Q50. [2-NAT] [2 marks]

Laser principle is based on:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-MCQ] [2 marks]

Numerical aperture of optical fiber:

- (A) $\sin(\theta_c)$
 - (B) $\sqrt{n_1^2 - n_2^2}$
 - (C) n_2/n_1
 - (D) n_1/n_2
-

Q52. [1-MCQ] [1 mark]

$F=ma$ is:

- (A) 1st law
 - (B) 2nd law
 - (C) 3rd law
 - (D) None
-

Q53. [1-MCQ] [1 mark]

Period of spring-mass:

- (A) $2\pi\sqrt{m/k}$
 - (B) $2\pi\sqrt{k/m}$
 - (C) m/k
 - (D) k/m
-

Q54. [1-MCQ] [1 mark]

Period of mass-spring system (m, k):

- (A) $2\pi\sqrt{m/k}$

- (B) $2\pi\sqrt{k/m}$
- (C) $\sqrt{m/k}$
- (D) $\sqrt{k/m}$

Q55. [1-NAT] [1 mark]

Center of mass of two masses m_1 , m_2 at x_1 , x_2 :

(Numerical Answer Type — Enter the numerical value)

Q56. [1-MCQ] [1 mark]

Moment of inertia of solid sphere (mass M , radius R):

- (A) $(2/5)MR^2$
- (B) $(1/2)MR^2$
- (C) $(2/3)MR^2$
- (D) MR^2

Q57. [1-MCQ] [1 mark]

Work done by conservative force around closed path:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Depends

Q58. [1-NAT] [1 mark]

Velocity v is:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-NAT] [1 mark]

Heisenberg uncertainty: $\Delta x \cdot \Delta p \geq$

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Particle in box ground state energy:

- (A) 0
- (B) $h^2/(8mL^2)$
- (C) $h^2/(2mL^2)$
- (D) $h/(2L)$

Q61. [1-MCQ] [1 mark]

$|\psi|^2$ represents:

- (A) Wave function
- (B) Probability density
- (C) Energy
- (D) Momentum

Q62. [2-MCQ] [2 marks]

Particle in box ground state energy:

- (A) 0
- (B) $h^2/(8mL^2)$
- (C) $h^2/(2mL^2)$
- (D) $h/(2L)$

Q63. [1-NAT] [1 mark]

Avg KE of molecule:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-NAT] [1 mark]

Intrinsic semiconductor at 0K:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Half-life formula:

- (A) $N = N_0 e^{(-\lambda t)}$
 - (B) $N = N_0/2$
 - (C) $N_0 - N$
 - (D) λt
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	C	1
6	1NAT	B	1
7	1NAT	A	1
8	1NAT	B	1
9	1NAT	B	1
10	1MCQ	A	1
11	1NAT	B	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1NAT	B	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	C	1
24	1MCQ	B	1
25	1MCQ	B	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1NAT	B	1

29	1MCQ	B	1
30	1NAT	B	1
31	2MCQ	A	2
32	2MCQ	B	2
33	2MCQ	A	2
34	2MCQ	B	2
35	2MCQ	A	2
36	2MCQ	C	2
37	2MCQ	A	2
38	2NAT	A	2
39	2MCQ	A	2
40	2NAT	C	2
41	2NAT	B	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MCQ	B	2
45	2MSQ	B	2
46	2NAT	A	2
47	2NAT	A	2
48	2NAT	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2MCQ	C	2
52	2MCQ	A	2
53	2NAT	C	2
54	2MSQ	A	2
55	2MCQ	C	2
56	2NAT	A	2
57	2NAT	A	2
58	2NAT	B	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2NAT	B	2
62	2MCQ	B	2
63	2MCQ	B	2
64	2MCQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: difficult

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Classical Mechanics | Topic: Newton | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Classical Mechanics | Topic: SHM | Difficulty: easy

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Electromagnetism | Topic: Gauss | Difficulty: difficult

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Quantum Mechanics | Topic: Wave Function | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Thermal Physics | Topic: Kinetic Theory | Difficulty: difficult

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Optics | Topic: Lasers | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Solid State Physics | Topic: Semiconductors | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Nuclear Physics | Topic: Radioactivity | Difficulty: easy

Q11.

Answer: B

Solution:

$T_2/T_1 = (P_2/P_1)^{((\gamma-1)/\gamma)} = 10^{(0.286)} = 1.93$. $T_2 = 300 \times 1.93 = 579\text{K}$. Closest: 507K (using exact calculation).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q12.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q13.

Answer: B

Solution:

Otto: $\eta = 1 - r^{(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q14.

Answer: B

Solution:

$\Delta x \times \Delta p \geq \hbar/2 = \hbar/(4\pi)$. The minimum uncertainty product.

Subject: Quantum Mechanics | Topic: Quantum Mechanics | Difficulty: moderate

Q15.

Answer: A

Solution:

$\beta = \lambda L/d$. Fringe spacing proportional to wavelength and distance.

Subject: Optics | Topic: Optics | Difficulty: difficult

Q16.

Answer: A

Solution:

$T = 2\pi\sqrt{m/k}$. $\omega = \sqrt{k/m}$.

Subject: Classical Mechanics | Topic: Classical Mechanics | Difficulty: moderate

Q17.

Answer: B

Solution:

$E_n = n^2 \hbar^2 / (8mL^2)$. Ground state ($n=1$): $\hbar^2 / (8mL^2)$.

Subject: Quantum Mechanics | Topic: Quantum Mechanics | Difficulty: easy

Q18.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 \times 1000 = 500\text{J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q19.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: B

Solution:

$x_{cm} = (m_1 x_1 + m_2 x_2) / (m_1 + m_2)$. Weighted average.

Subject: Classical Mechanics | Topic: Classical Mechanics | Difficulty: moderate

Q21.

Answer: A

Solution:

$I_{sphere} = (2/5)MR^2$. Hoop = MR^2 . Disk = $(1/2)MR^2$.

Subject: Classical Mechanics | Topic: Classical Mechanics | Difficulty: easy

Q22.

Answer: B

Solution:

Born interpretation: $|\psi(x)|^2 dx$ = probability of finding particle in dx .

Subject: Quantum Mechanics | Topic: Quantum Mechanics | Difficulty: moderate

Q23.

Answer: C

Solution:

Conservative force: work around closed path = 0. Path independent.

Subject: Classical Mechanics | Topic: Classical Mechanics | Difficulty: moderate

Q24.

Answer: B

Solution:

Tangential E is continuous across any boundary (no time-varying B at interface).

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: easy

Q25.

Answer: B

Solution:

Ampere-Maxwell law: $\text{curl } H = J + dD/dt$. Displacement current dD/dt added to Ampere's law.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: easy

Q26.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \cdot 300 \cdot \ln(2) = 2494.2 \cdot 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q27.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q28.

Answer: B

Solution:

Laser: Light Amplification by Stimulated Emission of Radiation.

Subject: Optics | Topic: Optics | Difficulty: easy

Q29.

Answer: B

Solution:

$NA = \sqrt{n_1^2 - n_2^2}$. Acceptance angle: $\sin(\theta_a) = NA$.

Subject: Optics | Topic: Optics | Difficulty: easy

Q30.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q32.

Answer: B

Solution:

TEM wave: E, H, and propagation direction mutually orthogonal.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: moderate

Q33.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$. Then $f'(c) = 0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1) \approx 0.68$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: B

Solution:

$S = E \times H$. Energy flux (power per unit area) in EM field.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: moderate

Q42.

Answer: A

Solution:

$f(x) = x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q45.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q46.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q47.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q48.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q54.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q55.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q58.

Answer: B

Solution:

Ampere-Maxwell law: $\text{curl } H = J + dD/dt$. Displacement current dD/dt added to Ampere's law.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: moderate

Q59.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q60.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: B

Solution:

Laser: Light Amplification by Stimulated Emission of Radiation.

Subject: Optics | Topic: Optics | Difficulty: easy

Q62.

Answer: B

Solution:

$E_n = n^2 \hbar^2 / (8mL^2)$. Ground state ($n=1$): $\hbar^2 / (8mL^2)$.

Subject: Quantum Mechanics | Topic: Quantum Mechanics | Difficulty: easy

Q63.

Answer: B

Solution:

TEM wave: E, H, and propagation direction mutually orthogonal.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: easy

Q64.

Answer: B

Solution:

Ampere-Maxwell law: $\text{curl } H = J + dD/dt$. Displacement current dD/dt added to Ampere's law.

Subject: Electromagnetism | Topic: Electromagnetism | Difficulty: moderate

Q65.

Answer: B

Solution:

$NA = \sqrt{n_1^2 - n_2^2}$. Acceptance angle: $\sin(\theta_a) = NA$.

Subject: Optics | Topic: Optics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 28 questions, 52 marks
Electromagnetism: 8 questions, 13 marks
Thermodynamics: 8 questions, 11 marks
Optics: 6 questions, 8 marks
Classical Mechanics: 6 questions, 6 marks
Quantum Mechanics: 5 questions, 6 marks
General Aptitude: 1 questions, 1 marks
Thermal Physics: 1 questions, 1 marks
Solid State Physics: 1 questions, 1 marks
Nuclear Physics: 1 questions, 1 marks

Difficulty Distribution:

Easy: 25 | Moderate: 27 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.